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Q1.Load the Heart Disease dataset using pandas and display the first 10 rows.

Also check the shape and basic information (info()) of the dataset.

```
import pandas as pd
df = pd.read_csv('heart.csv')
print("First 10 rows of the dataset:")
print(df.head(10))
print("\nShape of the dataset:")
print(df.shape)
print("\nDataset Information:")
print(df.info())
```

Q2.Check for missing values in the dataset. Show the count of null values for each column.

```
import pandas as pd
df = pd.read_csv('heart.csv')
print("Missing values in each column:")
print(df.isnull().sum())
```

Q3.Check for duplicate rows in the dataset. If any duplicates are found, remove

them and print the new shape of the dataset.

```
import pandas as pd

df = pd.read_csv('heart.csv')

duplicate_count = df.duplicated().sum()

print("Number of duplicate rows:", duplicate_count)

if duplicate_count > 0:

    df = df.drop_duplicates()

    print("Duplicate rows removed.")

print("New shape of the dataset:", df.shape)
```

Q4. Identify unrealistic/invalid values:

Count how many rows have Cholesterol = 0.

Count how many rows have RestingBP = 0.

Print both counts.

```
import pandas as pd

df = pd.read_csv('heart.csv')

cholesterol_zero_count = (df['Cholesterol'] == 0).sum()

restingbp_zero_count = (df['RestingBP'] == 0).sum()

print("Number of rows with Cholesterol = 0:", cholesterol_zero_count)

print("Number of rows with RestingBP = 0:", restingbp_zero_count)
```

Q5. Clean the invalid values:

Replace Cholesterol = 0 with the mean cholesterol value (excluding zeros).

Replace RestingBP = 0 with the mean resting blood pressure value (excluding zeros).

Round both columns to 2 decimal places.

Print the statistical summary (describe()) of these two columns before and after cleaning.

```
import pandas as pd
```

```
import numpy as np
```

```
df = pd.read_csv('heart.csv')
```

```
print("Statistical Summary Before Cleaning:")
```

```
print(df[['Cholesterol', 'RestingBP']].describe())
```

```
mean_cholesterol = df[df['Cholesterol'] != 0]['Cholesterol'].mean()
```

```
mean_restingbp = df[df['RestingBP'] != 0]['RestingBP'].mean()
```

```
df['Cholesterol'] = df['Cholesterol'].replace(0, mean_cholesterol)
```

```
df['RestingBP'] = df['RestingBP'].replace(0, mean_restingbp)
```

```
df['Cholesterol'] = df['Cholesterol'].round(2)
```

```
df['RestingBP'] = df['RestingBP'].round(2)
```

```
print("\nStatistical Summary After Cleaning:")
```

```
print(df[['Cholesterol', 'RestingBP']].describe())
```

Q6.Create a function to plot histograms for the following numerical columns:

Age

RestingBP

Cholesterol

MaxHR

Plot all four histograms in one figure using subplots (2x2 layout). Use this function to visualize the data after cleaning.

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

df = pd.read_csv('heart.csv')

mean_chol = df[df['Cholesterol'] != 0]['Cholesterol'].mean()

mean_bp = df[df['RestingBP'] != 0]['RestingBP'].mean()

df['Cholesterol'] = df['Cholesterol'].replace(0, mean_chol)

df['RestingBP'] = df['RestingBP'].replace(0, mean_bp)

df['Cholesterol'] = df['Cholesterol'].round(2)

df['RestingBP'] = df['RestingBP'].round(2)

def plot_histogram(ax, data, column_name):

    ax.hist(data[column_name], bins=20, edgecolor='black')

    ax.set_title(f'Histogram of {column_name}')

    ax.set_xlabel(column_name)

    ax.set_ylabel('Frequency')

fig, axes = plt.subplots(2, 2, figsize=(12, 8))

plot_histogram(axes[0, 0], df, 'Age')

plot_histogram(axes[0, 1], df, 'RestingBP')

plot_histogram(axes[1, 0], df, 'Cholesterol')

plot_histogram(axes[1, 1], df, 'MaxHR')

plt.tight_layout()

plt.show()
```

Q7. Identify and print numerical columns and categorical columns separately.

```
import pandas as pd

df = pd.read_csv('heart.csv')

numerical_columns = df.select_dtypes(include=['int64', 'float64']).columns.tolist()

categorical_columns = df.select_dtypes(include=['object']).columns.tolist()

print("Numerical Columns:")

print(numerical_columns)

print("\nCategorical Columns:")

print(categorical_columns)
```

Q8. Perform One-Hot Encoding on all categorical columns using

`pd.get_dummies()`.

Store the result in a new dataframe called `df_encoded`.

Print the shape and the first 5 rows of the encoded dataframe.

```
import pandas as pd

df = pd.read_csv('heart.csv')

categorical_columns = df.select_dtypes(include=['object']).columns

df_encoded = pd.get_dummies(
    df,
    columns=categorical_columns,
    drop_first=False
```

```
)
print("Shape of Encoded DataFrame:")
print(df_encoded.shape)
print("\nFirst 5 Rows of Encoded DataFrame:")
print(df_encoded.head())
```

Q9.After cleaning and encoding:

Print the final shape of df_encoded.

Show the list of all column names in the final dataframe.

```
import pandas as pd
import numpy as np

df = pd.read_csv('heart.csv')
mean_chol = df[df['Cholesterol'] != 0]['Cholesterol'].mean()
mean_bp = df[df['RestingBP'] != 0]['RestingBP'].mean()
df['Cholesterol'] = df['Cholesterol'].replace(0, mean_chol)
df['RestingBP'] = df['RestingBP'].replace(0, mean_bp)
df['Cholesterol'] = df['Cholesterol'].round(2)
df['RestingBP'] = df['RestingBP'].round(2)
categorical_columns = df.select_dtypes(include=['object']).columns
df_encoded = pd.get_dummies(
    df,
    columns=categorical_columns,
    drop_first=False
```

```
)  
print("Final Shape of df_encoded:")  
print(df_encoded.shape)  
print("\nColumn Names in Final DataFrame:")  
print(df_encoded.columns.tolist())
```

Q10.(Summary)

Write a short summary (in comments or markdown) covering:

1. What invalid values did you find and how did you fix them?
2. Why is it important to handle invalid values like Cholesterol = 0?
3. What is the purpose of One-Hot Encoding?
4. Any other observations from the cleaning process.